

		American International University-Bangladesh Faculty of Engineering Department of Electrical and Electronic Engineering			
Course Name:	Microprocessor and Embedded Systems			Course Code:	EEE 4103
Semester:	Summer 2025-26	Term:	Mid	Section:	
Faculty Name:				Assignment #:	01

Course Outcome Mapping with Questions

Item	COs	POIs	K	P	A	Marks	Obtained Marks
Q1	CO2	P.a.4.C3	K4	P1, P3, P7		10	
Q2-6	CO1	P.a.4.C3	K4	P1, P3, P7		2×5	
Total:						20	

Student Information:

Due Date:	16 July 2026										Submission Date:						Group #:	
Sl. #	Student ID #										Student Name	Department	Signature					
	p	q	-	a	b	c	d	e	-	r								

Instructions:

Please write your answers on A4-sized white paper. Then, take pictures of them or scan them and upload the files in .pdf format in TEAMS. If you prefer, you can also submit a .docx file. Make sure to write your name, ID, and section on the top page. Save the file with your lab group number and section as the filename (such as, MES_K_G4.docx) before uploading. Finally, submit a hard copy to my office.

Marking Rubrics (to be filled by Faculty):

	Excellent [26-30]	Proficient [21-26]	Good [16-20]	Acceptable [7-15]	Unacceptable [1-6]	No Response [0]	
Problem #	Detailed unique response explaining the concept properly, and the answer is correct with all works clearly shown.	Responses with no apparent errors, and the answer is correct, but the explanation is not adequate/unique.	The response shows an understanding of the problem, but the final answer may not be correct	Partial problem is solved; the response indicates that part of the problem was not understood clearly or not solved.	Unable to clarify the understanding of the problem, and the method of problem-solving was not correct	No Response/ copied from others/identical submissions with gross errors/image file printed	Secured Marks
Comments						Total Marks (30)	

Question # 1: Complete Table 1 after reviewing the specified microcontrollers' datasheet.

[10]

Table 1

Specifications	ATMega328PB	STM32F401RD	STM32F469AE	ATMega2561	STM32F446xC/E
Architecture Type					
Maximum Clock Speed					
Program Flash Memory (KB)					
SRAM (KB)					
EEPROM (KB)					
ADC Resolution					
Operating Voltage Range (V)					
Number of Timers					
Number of PWM Channels					
Communication Interfaces					

The unit prices of the above-mentioned MCUs are as follows: (1 USD = 121 BDT)

	ATMega328PB	STM32F401RD	STM32F469AE	ATMega2561	STM32F446xC/E
Price	\$1.20	\$6.05	\$6.85	\$10.3	\$3.95

ABCD Industry in Bangladesh is trying to develop an affordable shop security system, and they have shortlisted the 5000 MCUs listed as possible candidates for their system CPU. The required minimum specifications for their intended design for the CPU are given below:

System Parameters	Minimum Requirements
Minimum Clock Speed	16 MHz
Minimum SRAM	16 KB
ADC Resolution	12-bit
Program Memory	64 KB
Number of PWM Channels	15
Number of Timers	15
Serial Communication Interfaces	4 SPIs, 3 TWIs, 4 USARTs

Being a design engineer at *ABCD* Industry, you have been given the responsibility of selecting the most suitable IC from the list for the security system design that requires a minimum of 5000 microcontroller ICs. **Select** an appropriate IC from the list to design an affordable and efficient system and justify your answer with proper reasoning. **Compute** the total cost of designing the system.

[Until and unless specified, the digits from your ID will sit side-by-side, no multiplication among them! Total Marks: 5×4 = 20]

1. In a biscuit factory, a control system has been designed based on an Arduino Uno microcontroller. This system transfers the biscuit packets at a regular time interval to the conveyor belt from the biscuit-making machine. It was found that the Timer modules in the Arduino Uno are perfect candidates for counting the number of biscuit packets being transferred to the conveyor belt correctly and appropriately. Compute the number of biscuit packets being transferred using the Timers within the following time intervals (Timer0 or Timer1 may be used as needed).
 - a) $r0$ ms
 - b) abc ms

Assume that the Arduino system clock has been set at $1q$ MHz and the available pre-scalers of the system are 1, 8, 64, 256, 512, and 1024. Comment on why and how the required counts are possible to implement using Timer0 or Timer1. Write down the program codes for the Arduino board. Show the simulation results using Proteus.

2. (a) A TV remote controller is to be designed for the selection of channels. The channel selection buttons are required to be debounced, that is, one press must cause the remote to change to the subsequent channel. It was observed that the switches exhibit bounce times well under b ms and the duty cycle is 50%. Design a circuit using the 74HC14 Schmitt trigger IC along with the resistance and capacitance. The worst-case V_{th} of 74HC14 for a signal going LOW is $2.r$ V and that of when going high is $0.pq$ V. Also, consider that the CMOS device leakage current is pa μ A and the gate's best-case switching point is of the order of $0.r$ V. What is the hysteresis voltage?

(b) Write a software code for the Arduino to read the status of a switch connected to pin number p by avoiding the bouncing problem of the switch and then display that switch status in the LED connected to pin number $1r$. If the switch status is HIGH/LOW, the LED will be turned ON/OFF respectively. It was observed that the switches exhibit bounce times well under $q0$ ms and the duty cycle is 50%. Use the Timer0 or Timer1 function of the Arduino to compute this delay. Draw the flow chart of this program. Show the simulation results using Proteus.

3. A biomedical engineer is designing a new laboratory where biologically hazardous materials are used for experimental research works. As such, during the experimental works are ongoing, no one is allowed to remain inside the laboratory. To display the message that hazardous materials are being used in the laboratory, 2 lights with green and red colors are installed at the entrance door of the laboratory. When the green light is turned on, it indicates that anybody can enter the laboratory, but when the red light is turned on then no one can enter the laboratory. It is to be ensured that both lights can't remain turned on at the same time and these 2 lights should be controlled using a single mechanical switch inside the laboratory, and they can only light up alternatively when the switch is pressed or depressed. The switch has already been debounced using a hardware circuit. Draw a flowchart to show the logical flow of the program and then write a program for the Arduino Uno to operate the door lights based on the switch conditions as per the given constraints. Show the simulation results using Proteus.

4. (a) Write an Arduino microcontroller code for an acd ms interval LED blinking system using Timer x ($x = 0-2$) interrupts and a pre-scalar value of 1024. LED is connected to the pin r of the board. Compute the value to be loaded into the OCR($r-1$)A register. Show the simulation results using Proteus.

(b) Use 2 pins in the same port for pin change interrupt. Use Arduino Pin6 and Pin7, which maps to (PD6 and PD7 in PORTD). The 2 PCINT input pins must be connected to 2 push buttons. Use the interrupt events to toggle 2 LEDs, connected to the pin $pin2$ and $pin3$ (mapped to PD2 and PD3 of PORTD) in the respective ISR handler function for each PCINT interrupt. Draw the flow chart of this program. Show the simulation results using Proteus.

(c) Use Arduino Pin2, which maps to (PD2 in PORTD) for External Interrupt Request, INT0. Use Arduino Pinp and Pinr, which are mapped to corresponding port pins in PORTB to toggle 2 LEDs, connected to the pinp and pinr, respectively. The INT0 external interrupt request input pin, PD2 (PORTD) must be connected to a push button. Use the interrupt events in the respective ISR handler function for the INT0 interrupt. Draw the program flow chart. Show the Proteus simulation results.

5. (a) Determine the output of the following program. Compute the duration that Arduino blinks an LED and makes the system power down. When powered down, the Arduino's current drops from 9a.d mA to just ab mA. The supply voltage is 5.0abc V. Compute the amount of power that is saved by the Arduino during this power-down mode.

```
#include "LowPower.h"
#define RED_PIN a

int red_on = pqabc;
int red_off = qabd

void setup() {
  pinMode(a, OUTPUT);
}

void loop() {
  digitalWrite(a, HIGH);
  delay(red_on);
  digitalWrite(a, LOW);
  delay(red_off);
  LowPower.powerDown(SLEEP_4S, WDT_OFF, BOD_OFF, SPI_OFF, TWI_OFF);
}
```

- (b) Write the output of the following program. Compute power and energy consumption during the ON and OFF conditions as well as SLEEP modes in idle and power down states. Compute power savings.

```
#include "LowPower.h"

#define RED_PIN c

int red_on = rpace
int red_off = qade

void loop() {
  pinMode(c, OUTPUT);
}

void setup() {
  digitalWrite(c, HIGH);
  delay(red_on); // USB Meter readings = 4.9ac V, 0.7der A)
  digitalWrite(c, LOW);
  delay(red_off); // USB Meter readings = 5.0bcd V, 0.rpq A)
  LowPower.idle(SLEEP_8S, ADC_OFF, SPI_OFF); // USB Meter readings = 5.ab V, 0.00rd A)
  LowPower.powerDown(SLEEP2S, WDT_OFF, BOD_OFF); // USB Meter readings = 5.0cd V, 0.00ae A)
}
```